



IC-AITEWA
2027

INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE
AND INTELLIGENT TECHNOLOGIES FOR
ENERGY, WATER AND AUTOMATION
TKM COLLEGE OF ENGINEERING, KOLLAM, INDIA

Department of Mechanical Engineering
TKM College of Engineering, Kollam, Kerala, India
(A Govt. Aided and Autonomous Institution)

International Conference on Artificial Intelligence and Intelligent Technologies for Energy, Water and Automation (IC-AITEWA 2027)

"Advancing Sustainable Energy, Water Security and Smart
Automation through Intelligent Technologies"

March 18-20
2027

About

TKM College of Engineering Kollam, Kerala, India

Established in 1958 by the visionary reformer and philanthropist Janab Thangal Kunju Musaliar under the aegis of the TKM College Trust, TKM College of Engineering (TKMCE) in Kollam is the oldest engineering college in the private sector in Kerala. Over nearly seven decades, the institution has stood as a premier torchbearer for technological and societal transformation, producing thousands of globally recognized engineers and industry stalwarts.

Headed by the TKM College Trust and currently led by Janab Shahal Hassan Musaliar, the autonomous institution is affiliated with APJ Abdul Kalam Technological University. Operating with eleven academic departments, TKMCE offers comprehensive Undergraduate, Postgraduate, Doctoral programmes and Post Doctoral programmes. All eligible B.Tech programmes are accredited by the National Board of Accreditation (NBA), reflecting the campus's unwavering commitment to academic rigor, state-of-the-art research facilities, and engineering excellence. Housed within a distinctive architectural landmark, the campus continues to foster innovation on a national and international scale.

Department of Mechanical Engineering

The Department of Mechanical Engineering is one of the pioneering branches of TKM College of Engineering. Established in 1958, it stands as one of the oldest and largest departments offering a degree in Mechanical Engineering in Kerala, comprising 46 faculty members and 33 technical staff. Driven by the vision of "Excellence in Mechanical Engineering with a perspective of sustainable development," the department offers premier, NBA-accredited Undergraduate (B.Tech), Postgraduate (M.Tech), Doctoral (Ph.D.) programmes and Post Doctoral programmes (PDF). As an AICTE-approved QIP and ADF research center affiliated with APJ Abdul Kalam Technological University and the University of Kerala, it routinely secures funding from top national bodies including ISRO, DAE, ARDB, and ANERT.

Key Research Frontiers: Advanced Materials, Automation, Biomechanics, Cryogenics, Fracture Mechanics, HVAC & Refrigeration, Nanomaterials & Nanofluids, Solar Energy Systems, and Superconductivity.

About

IC-AITEWA 2027

IC-AITEWA 2027 — International Conference on Artificial Intelligence and Intelligent Technologies for Energy, Water and Automation brings together researchers, academicians, industry leaders, innovators, and young researchers to explore how intelligent technologies can address some of the world's most pressing engineering challenges. Held at TKM College of Engineering, Kollam, the three-day international conference is centred on the theme "Advancing Sustainable Energy, Water Security and Smart Automation through Intelligent Technologies." The conference will create a dynamic platform for exchanging ideas, presenting cutting-edge research, fostering interdisciplinary collaboration, and building meaningful international and industry partnerships across **Artificial Intelligence, Sustainable Energy, Water Technologies, Intelligent Manufacturing, Robotics, Digital Twins, Industry 5.0, and Smart Infrastructure**. Featuring international and national plenary keynote speakers, technical sessions, expert panels, research presentations, innovation showcases, pre-conference workshops, and an industry exhibition, IC-AITEWA 2027 aims to connect research with real-world impact and bring together diverse perspectives to shape a smarter, more sustainable, and technologically advanced future.

PATRONS AND ORGANIZING TEAM

Chief Patron



Sri. T. K. Shahal Hassan Musaliar
President, TKM College Trust

Honorary Patron



Dr. S. Somanath
Former Chairman of ISRO,
Distinguished Alumni of
TKMCE

Patrons



Sri. T. K. Jalaluddin Musaliar
Treasurer, TKM College
Trust



Sri. Jamaludheen Musaliyar
Member, TKM College Trust

Conference Chair



Dr. Sadiq A.
Principal, TKMCE

Conference Co-Chair



Dr. Sudheer A.
Professor & Head,
Department of
Mechanical
Engineering

Organizing Secretaries



Dr. Takashi Suzuki
Professor, Sophia
University, Japan



Dr. Jan Kazak
Associate Professor,
Wroclaw University of
Environmental and Life
Sciences, Poland

Organizing Secretaries



Dr. Baiju V.
Associate Professor,
Department of
Mechanical
Engineering



Dr. Jesna Mohammed
Associate Professor,
Department of
Mechanical
Engineering



Dr. Chai Ai Bao
Associate Professor,
University of
Nottingham,
Malaysia



Dr. Mohamed R. Ali
Professor, Sohar
University, Oman

Joint Secretaries

Dr. Leena R.

Assistant Professor, Department of Mechanical Engineering

Dr. Firoz N.

Assistant Professor, Department of Mechanical Engineering

Dr. Faraz P. Junaid

Assistant Professor, Department of Mechanical Engineering

Prof. Rizwan Rasheed

Assistant Professor, Department of Mechanical Engineering

Dr. Anand Sekhar R.

Assistant Professor, Department of Mechanical Engineering

Prof. Ahammed Bilal

Assistant Professor, Department of Mechanical Engineering

Prof. P. G. Karthik

Assistant Professor, Department of Mechanical Engineering

Prof. Mohammed Irfan A.

Assistant Professor, Department of Mechanical Engineering

Advisory Committee

International Advisory Committee

1. **Dr. Mitsuhsa Ichiyanagi**, Sophia University, Japan
2. **Dr. K. Pratheesh**, University of Technology and Applied Science, Muscat, Sultanate of Oman
3. **Dr. Nibal Fadel Farman Alhialy**, University of Baghdad, Iraq
4. **Dr Sathish Clastinrusselraj Indirathankam**, University of Newcastle, Australia
5. **Dr. Sandra Boetcher**, Embry– Riddle Aeronautical University, USA
6. **Dr. Afrah Turki Awad**, Northern Technical University, Iraq
7. **Dr. Aravind P. V.**, University of Groningen, Netherlands
8. **Prof. Dr Teo Fang Yenn**, University of Nottingham, Malaysia
9. **Prof. Maurice Collins**, University of Limerick, Ireland
10. **Prof. Anthony (Tony) James Robinson**, University of Dublin, Trinity College, Ireland.
11. **Sri. Roopesh Kaimal**, Manager Thermal and Aero, CAE Vinfast Auto Ltd, Vietnam
12. **Sri. Kalidas Pavithran**, Director – Project Controls, Drydocks world Dubai, UAE
13. **Vijay Krishna C. V. ,** Marketing Lead , Scale Ireland , Ireland

National Advisory Committee

1. **Sri. Jayakrishnan S.**, Chief Executive officer(Bangalore Complex), Hindustan Aeronautics Limited (HAL), Bengaluru
2. **Dr. Dibakar Rakshit**, Indian Institute of Technology Delhi
3. **Dr. Krishna Anand V. G.**, Ramaiah University of Applied Sciences, Bengaluru
4. **Dr. Rajesh Baby**, St. Joseph's College of Engineering and Technology, Palai
5. **Dr. Ravish G.**, Mythri Aqua Tech, Hyderabad
6. **Dr. Joy Varghese V. M.**, College of Engineering, Thiruvananthapuram
7. **Dr. Anish K. John**, Govt Engineering College Barton hill, Thiruvananthapuram
8. **Dr. Pallab Sinha Mahapatra**, Indian Institute of Technology Madras
9. **Dr. Vinay V. Panicker**, National Institute of Technology Calicut, Kozhikode
10. **Dr. Ravita Lamba**, Indian Institute of Technology Roorkee, Uttarakhand
11. **Dr. Sanjay Kumar Singh**, Indian Institute of Technology (BHU), Varanasi
12. **Dr. Sarvoththama Jothi**, National Institute of Technology Calicut, Kozhikode
13. **Dr. Ranjith M.**, National Institute of Technology Surathkal, Managalore
14. **Dr. Manikandan Sundararaj**, SRM Institute of Science & Technology, Chennai
15. **Dr. Anoop Velayudhan**, Indian Council of Medical Research (ICMR), New Delhi
16. **Dr. Salih A.**, Indian Institute of Space Science and Technology (IIST)
Thiruvananthapuram
17. **Dr. Jinshah B. S.**, Lead Engineer, GE Appliances-Haier, Hyderabad
18. **Reji Ravindran**, Group Manager, Power Train Installation, Vehicle Engineering,
Volvo Group India Pvt Ltd
19. **Alex James**, Professor and Controller of Examination, Digital University Kerala

Student Council

Parthiv J. Prasad	Panel Head, Department of Mechanical Engineering
Pranav R.	Panel Head, Department of Mechanical Engineering
Shrevan S. Nair	Panel Head, Department of Mechanical Engineering
Sathyajith N. S.	Tech Head, Department of Computer Science & Engineering
Amrutha R.	Student Coordinator, Department of Mechanical Engineering
Rohit Mangalassery	Student Coordinator, Department of Mechanical Engineering
Afnan N.	Student Coordinator, Department of Mechanical Engineering
Parthiv P.	Tech Coordinator, Department of Computer Science & Engineering
Rasal Musthafa	Tech Coordinator, Department of Computer Science & Engineering
Fathima Abdulla	Tech Coordinator, Department of Electrical & Computer Engineering
Ashna Maria Martin	Research Scholars, Department of Mechanical Engineering
Ajmi Midhlaj	Research Scholars, Department of Mechanical Engineering
Vishnu V. S.	Research Scholars, Department of Mechanical Engineering

CONFERENCE TECHNICAL TRACKS

Key Topics covered in the Conference, but not limited to:

1. Artificial Intelligence and Machine Learning

Deep learning architectures, Neural network applications in physical systems, IoT, AI in healthcare engineering, Smart city data analytics, Predictive computational modeling

2. Sustainable Energy Technologies

Renewable energy grid integration, Smart grid technologies, Hydrogen energy systems, Carbon-neutral innovations, Solar and thermal energy conversion, Decentralized power systems, Solar cooling, heating, power generation

3. Water Resources and Desalination

Advanced desalination processes, Industrial wastewater treatment, Water distribution network optimization, Smart water security solutions, Intelligent hydrological monitoring

4. Intelligent Manufacturing and Automation

Robotics and autonomous systems, Cyber-physical systems, Digital twin implementations, Smart factory automation, Industry 5.0 strategies, Modern manufacturing paradigms

5. Smart Materials and Advanced Engineering

Synthesis of smart and adaptive materials, Advanced structural engineering, Computational solid mechanics, Micro- and nano-materials, Sustainable building systems, composites, alloys, materials characteristics

SCHEDULE

NOV 30, 2026

EXTENDED ABSTRACT SUBMISSION

DEC 10, 2026

NOTIFICATION OF ACCEPTANCE

JAN 10, 2027

CAMERA READY PAPER – Final Paper

JAN 20, 2027

EARLY BIRD REGISTRATION

MAR 18 – 20, 2027

CONFERENCE

Accepted papers will be published in peer-reviewed journals and conference proceedings.



IC-AITEWA 2027

INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE
AND INTELLIGENT TECHNOLOGIES FOR
ENERGY, WATER AND AUTOMATION

— TKM COLLEGE OF ENGINEERING, KOLLAM, INDIA —

CONTACT DETAILS

For Enquiries :

Dr. Baiju V.

Associate Professor, Department of Mechanical Engineering

baiju@tkmce.ac.in

+91 9895937476

Dr. Jesna Mohammed

Associate Professor, Department of Mechanical Engineering

jesnamohammed@tkmce.ac.in

+91 9895010550

Mail ID :

icaitewa27@tkmce.ac.in

Website Link :

<https://IC-AITEWA-2027.tkmce.ac.in>